Year 10 Engineering Technology - Wind Powered USB Phone Charger

Lesson plans: 60 lessons across 20 weeks

Lesson Plan Overview

These lesson plans support a 20-week Stage 5 Engineering Technology alternate-energy project. They assume three lessons per week and spread renewable energy theory, impellor design, technical graphics, construction, output testing, improvement and folio evidence across the full semester.

Assumption: three lessons per week across 20 weeks. Each week below gives a practical lesson flow that can be expanded or compressed to suit timetable interruptions.

Stage 5 Outcomes

Outcome	Official outcome wording used for this unit
EGT5-SAF-01	applies risk management and safe work practices in engineering contexts
EGT5-COM-01	communicates ideas, concepts and solutions for engineering practice
EGT5-EVL-01	investigates and evaluates engineering systems and solutions
EGT5-IVT-01	explains engineering practices and the influence of technologies used in engineering industries
EGT5-GRP-01	develops and applies technical graphics in engineering contexts
EGT5-MEA-01	applies mechanical analysis and practical testing to investigate engineering concepts
EGT5-USE-01	selects and applies materials for engineering projects
EGT5-ENV-01	analyses relationships between engineering design, production and sustainability

20-Week Lesson Sequence

Week	Focus	Lesson flow	Evidence checkpoint	Outcomes
1	Unit launch and alternative energy context	Lesson A: introduce or revise core theory. Lesson B: apply through design, planning, build or testing. Lesson C: document evidence, check	Safety notes, energy comparison and project brief	EGT5-SAF-01, EGT5-

		quality and respond to feedback. Focus detail: Renewable and non-renewable energy; project context; WHS for rotating parts and low-voltage testing.	response.	IVT-01, EGT5-COM-01
2	Wind turbine operation	Lesson A: introduce or revise core theory. Lesson B: apply through design, planning, build or testing. Lesson C: document evidence, check quality and respond to feedback. Focus detail: Energy transfer chain; turbine parts; generator output; charging and storage limits.	Energy pathway diagram and vocabulary notes.	EGT5-IVT-01, EGT5-MEA-01
3	Renewable energy advantages and limits	Lesson A: introduce or revise core theory. Lesson B: apply through design, planning, build or testing. Lesson C: document evidence, check quality and respond to feedback. Focus detail: Intermittency, storage, charging infrastructure, regional use and reliability issues.	Advantages/limitations table and short response.	EGT5-ENV-01, EGT5-EVL-01
4	Energy costing and lifecycle impact	Lesson A: introduce or revise core theory. Lesson B: apply through design, planning, build or testing. Lesson C: document evidence, check quality and respond to feedback. Focus detail: Cost of energy production; materials, recyclability, disposal and sustainability impacts.	Research notes and sustainability evidence.	EGT5-ENV-01, EGT5-IVT-01
5	Project brief and generator constraints	Lesson A: introduce or revise core theory. Lesson B: apply through design, planning, build or testing. Lesson C: document evidence, check quality and respond to feedback. Focus detail: USB generator unit, dimensions, box size limit, portability, compactness and safety risks.	Constraint sheet and risk list.	EGT5-SAF-01, EGT5-COM-01
6	Impellor research	Lesson A: introduce or revise core theory. Lesson B: apply through design, planning, build or testing. Lesson C: document evidence, check quality and respond to feedback. Focus detail: Blade material, blade angle, number of blades, wind-catcher options and efficiency variables.	Impellor research notes and labelled blade ideas.	EGT5-MEA-01, EGT5-USE-01
7	Mechanical transfer options	Lesson A: introduce or revise core theory. Lesson B: apply through design, planning, build or testing. Lesson C: document evidence, check quality and respond to feedback. Focus detail: Direct drive, gears/cogs, shafts, bearings, friction, alignment and rotation losses.	Mechanical option comparison and annotated sketch.	EGT5-MEA-01, EGT5-EVL-01
8	Test variables and design	Lesson A: introduce or revise core theory. Lesson B: apply through	Test-variable plan and teacher	EGT5-SAF-

	constraints	design, planning, build or testing. Lesson C: document evidence, check quality and respond to feedback. Focus detail: Wind source, output measurement, fair testing, guarded moving parts and teacher checkpoint.	check record.	01, EGT5-EVL-01
9	Concept sketching	Lesson A: introduce or revise core theory. Lesson B: apply through design, planning, build or testing. Lesson C: document evidence, check quality and respond to feedback. Focus detail: Frame, impellor, shaft support, generator location, USB module placement and housing.	Two or more concept sketches.	EGT5-GRP-01, EGT5-COM-01
10	Concept comparison	Lesson A: introduce or revise core theory. Lesson B: apply through design, planning, build or testing. Lesson C: document evidence, check quality and respond to feedback. Focus detail: Compare concepts using function, safety, compactness, efficiency, build quality and appearance.	Concept comparison table and selected direction.	EGT5-EVL-01, EGT5-COM-01
11	Final design selection and build planning	Lesson A: introduce or revise core theory. Lesson B: apply through design, planning, build or testing. Lesson C: document evidence, check quality and respond to feedback. Focus detail: Justification, materials list, tools, construction sequence and risk-control update.	Final design justification and build sequence.	EGT5-GRP-01, EGT5-SAF-01
12	Final drawing or CAD model	Lesson A: introduce or revise core theory. Lesson B: apply through design, planning, build or testing. Lesson C: document evidence, check quality and respond to feedback. Focus detail: Onshape or final drawing with dimensions, labels, assembly thinking and component locations.	Final drawing/CAD evidence.	EGT5-GRP-01, EGT5-COM-01
13	Frame and shaft-support construction	Lesson A: introduce or revise core theory. Lesson B: apply through design, planning, build or testing. Lesson C: document evidence, check quality and respond to feedback. Focus detail: Begin construction; frame accuracy; shaft support; generator location; guarded moving parts.	Build log and construction photos.	EGT5-SAF-01, EGT5-USE-01
14	Impellor attachment and mechanical alignment	Lesson A: introduce or revise core theory. Lesson B: apply through design, planning, build or testing. Lesson C: document evidence, check quality and respond to feedback. Focus detail: Spin test; bearing support; wobble reduction; friction reduction and quality checks.	Spin-test notes and mechanical fault log.	EGT5-MEA-01, EGT5-EVL-01

15	Electrical connection and safe output testing	Lesson A: introduce or revise core theory. Lesson B: apply through design, planning, build or testing. Lesson C: document evidence, check quality and respond to feedback. Focus detail: Connect generator/USB unit; teacher-check wiring; measure output with approved equipment.	Output test data and teacher check record.	EGT5-SAF-01, EGT5-MEA-01
16	Fault-finding and construction evidence	Lesson A: introduce or revise core theory. Lesson B: apply through design, planning, build or testing. Lesson C: document evidence, check quality and respond to feedback. Focus detail: Identify low output, poor alignment, unstable frame or blade problems; document fixes.	Fault log, photos and improvement decisions.	EGT5-EVL-01, EGT5-COM-01
17	Controlled improvements	Lesson A: introduce or revise core theory. Lesson B: apply through design, planning, build or testing. Lesson C: document evidence, check quality and respond to feedback. Focus detail: Change one variable: blade angle, drive ratio, alignment, frame stability or wind source.	Improvement plan and modified prototype evidence.	EGT5-EVL-01, EGT5-MEA-01
18	Retesting and performance comparison	Lesson A: introduce or revise core theory. Lesson B: apply through design, planning, build or testing. Lesson C: document evidence, check quality and respond to feedback. Focus detail: Retest improvements, compare results and complete final output testing.	Before/after data and final test table.	EGT5-EVL-01, EGT5-COM-01
19	Sustainability and portability evaluation	Lesson A: introduce or revise core theory. Lesson B: apply through design, planning, build or testing. Lesson C: document evidence, check quality and respond to feedback. Focus detail: Review portability, size limit, materials, waste, construction quality and real-world use.	Evaluation draft using test evidence.	EGT5-ENV-01, EGT5-EVL-01
20	Folio completion and submission	Lesson A: introduce or revise core theory. Lesson B: apply through design, planning, build or testing. Lesson C: document evidence, check quality and respond to feedback. Focus detail: Final testing photos, folio completion, final evaluation and submission checkpoint.	Completed folio/report and project reflection.	EGT5-COM-01, EGT5-EVL-01, EGT5-IVT-01

Core Resources and Adjustments

- Generator/USB units; bearings; gears/cogs; blade materials; frame materials; shafts; hand tools; low-voltage test equipment; guarded fan or teacher-approved wind source.

- Student scaffolds: renewable energy comparison, generator constraint sheet, concept sketch template, final drawing checklist, test table, fault log and evaluation prompts.
- Adjustments: labelled diagrams, vocabulary list with images, partially completed final drawing and teacher-approved simplified impellor or direct-drive design.
- Extension: compare direct drive and geared drive, test blade angle or blade number, or fabricate accurate parts using laser cutting or 3D printing where available.